ALAN MATTHEW KNEE

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RESEARCH INTERESTS

Gravitational-wave astrophysics with ground- and space-based laser interferometric detectors, gravitational-wave data analysis and detector characterization. Physics of black holes and neutron stars. Bayesian and frequentist statistical analysis. Design and implementation of high-throughput analysis pipelines for gravitational-wave data.

POSITIONS

Postdoctoral Research Fellow — Sep 2025 – present
University of Michigan, *Ann Arbor, MI, USA*

Graduate Research Assistant — Sep 2019 – Aug 2025
University of British Columbia, *Vancouver, BC, Canada*

EDUCATION

Ph.D. Astronomy, University of British Columbia — Sep 2021 – Aug 2025
Supervisor: Jess McIver
Thesis: *Searches for novel gravitational-wave sources with ground- and space-based detectors*

M.Sc. Astronomy, University of British Columbia — Sep 2019 – Aug 2021
Supervisor: Jess McIver
Thesis: *Prospects for measuring off-axis spins of binary black holes with Plus-era gravitational-wave detectors*

B.Sc. (Honours) Physics, University of British Columbia — Sep 2014 – May 2019
with Distinction

SCHOLARSHIPS & AWARDS

J. S. Plaskett Medal – for most outstanding PhD thesis in astronomy in Canada	2026
UBC Four Year Doctoral Fellowship	2022 – 2025
Killam Doctoral Scholarship	2022 – 2024
John I. Watters Research Fellowship	2021
NSERC Canada Graduate Scholarships – Master's	2019 – 2020
NSERC Undergraduate Summer Research Award (×2)	2017 & 2018

PRESENTATIONS – INVITED

1. University of British Columbia Gravity Seminar, *Vancouver, BC, Canada* — Sep 2026
2. University of Victoria Physics and Astronomy Colloquium, *Victoria, BC, Canada* — Sep 2026
3. NRC Herzberg Astronomy Colloquium, *Victoria, BC, Canada* — Sep 2026
4. CASCA Annual General Meeting Plaskett Medal Plenary, *Montréal, QC, Canada* — Jun 2026
5. University of Washington CENPA Seminar, *Seattle, WA, USA* — Jun 2025
6. University of Toronto Astrostats/ML Seminar, *Toronto, ON, Canada (virtual)* — Feb 2024
7. Joint Statistical Meetings, *Toronto, ON, Canada* — Aug 2023

PRESENTATIONS – CONTRIBUTED

1. **LVK Meeting**, *Pisa, Italy* — Mar 2026
   CW and Stochastic DetChar update and status
2. **GWPAW**, *Atlanta, GA, USA* — Dec 2025
   Constraints on gravitational waves from the 2024 Vela pulsar glitch
3. **GR24 & Amaldi16**, *Glasgow, UK* — Jul 2025
   Identifying loud gravitational-wave bursts from galactic black hole binaries with LISA
4. **Testing Gravity 2025**, *Vancouver, BC, Canada* — Jan 2025
   Searching for continuous gravitational waves with a hidden Markov model
5. **LVK Meeting**, *Barcelona, Spain* — Sep 2024
   CW and Stochastic DetChar update and status

6. **GWANW**, *LIGO Hanford, WA, USA* Jun 2024
   Introduction to LISA
7. **APS April Meeting**, *Sacramento, CA, USA* Apr 2024
   Identifying loud gravitational-wave bursts from galactic black hole binaries with LISA
8. **LVK Meeting**, *Baton Rouge, LA, USA* Mar 2024
   CW and Stochastic DetChar update and status
9. **Amaldi15**, *virtual* Jul 2023
   Hierarchical search for persistent gravitational-wave candidates in O3 LIGO data
10. **GWANW**, *LIGO Hanford, WA, USA* Jun 2023
    Introduction to GW parameter estimation
11. **APS April Meeting**, *Minneapolis, MN, USA* Apr 2023
    Characterizing LISA-band gravitational-wave signals from eccentric compact binaries orbiting a supermassive black hole
12. **LVK Meeting**, *Evanston, IL, USA* Mar 2023
    Hierarchical search for persistent gravitational waves from sub-threshold all-sky all-frequency candidates
13. **GWANW**, *LIGO Hanford, WA, USA* Jun 2022
    Introduction to GW parameter estimation
14. **APS April Meeting**, *New York, NY, USA (virtual)* Apr 2022
    Translating inferences between eccentric gravitational waveform models
15. **Amaldi14**, *virtual* Jul 2021
    Prospects for measuring off-axis spins of binary black hole sources with A+/AdV+
16. **CAP Congress**, *virtual* Jun 2021
    Prospects for measuring off-axis spins of binary black hole sources with A+/AdV+
17. **APS April Meeting**, *virtual* Apr 2021
    Prospects for measuring off-axis spins of binary black hole sources with A+

PUBLICATIONS – SHORT AUTHOR LIST

1. L. Mirasola, F. Amicucci, M. Carrio, D. H. T. Cheung, M. A. Ferrer, I. Prohens, E. Goetz, D. Keitel, **A. M. Knee**, P. Leaci, C. Palomba, K. Pham, O. J. Piccinni, M. Pitkin, K. Riles, G. Woan, Z. Zhang, M. Bejger, A. Calafat, R. Jaume, D. I. Jones, A. Krolak, I. La Rosa, A. Melatos, J. R. Mérou, A. Nemmani, B. Rajbhandari, C. Salvadore, A. M. Sintes, R. Tenorio, K. Wette, J. T. Whelan, T. Yamamoto, *The G347.3-0.5 outlier from O3: a follow-up case study for continuous gravitational wave candidates*, submitted to CQG
2. **A. M. Knee**, K. Riles, L. Sun, *Spotlight searches for continuous gravitational waves triggered on radiometer candidates in LIGO O4a data*, submitted to PRD
3. Y. Lecoeuche, J. McIver, R. Udall, **A. M. Knee**, K. Rink, S. Hourihane, S. J. Miller, K. Chatziioannou, T. J. Massinger, D. Davis, *Coalescing compact binary parameter estimation with gravitational waves in the presence of non-Gaussian transient noise*, CQG **43** 215014 (2026)
4. D. H. T. Cheung, K. Riles, R. Amengual, P. Baxi, A. Calafat, A. Effler, T. A. Ferreira, E. Goetz, T. Kimpson, D. Keitel, **A. M. Knee**, J.-R. Merou, Q. L. Nguyen, J. O'Leary, O. J. Piccinni, A. M. Sintes, K. Wette, *Search for continuous gravitational waves from neutron stars in five globular clusters in the first part of the fourth LIGO-Virgo-KAGRA observing run*, PRD **114** 062001 (2026)
5. Z. Xuan, S. Naoz, A. K. Y. Li, B. Kocsis, E. Petigura, **A. M. Knee**, J. McIver, K. Kremer, W. M. Farr, *Extracting astrophysical information of highly eccentric binaries in the millihertz gravitational wave band*, PRD **111** 043018 (2025)
6. **A. M. Knee**, J. McIver, S. Naoz, I. M. Romero-Shaw, B.-M. Hoang, E. Grishin, *Detecting gravitational-wave bursts from black hole binaries in the Galactic Center with LISA*, ApJL **971** L38 (2024)
7. **A. M. Knee**, H. Du, E. Goetz, J. McIver, J. B. Carlin, L. Sun, L. Dunn, L. Strang, H. Middleton, A. Melatos, *Search for continuous gravitational waves directed at subthreshold radiometer candidates in O3 LIGO data*, PRD **109** 062008 (2024)
8. N. Shah, **A. M. Knee**, J. McIver, D. C. Stenning, *Waves in a forest: a random forest classifier to distinguish between gravitational waves and detector glitches*, CQG **40** 235008 (2023)
9. **A. M. Knee**, I. M. Romero-Shaw, P. D. Lasky, J. McIver, E. Thrane, *A Rosetta Stone for eccentric gravitational waveform models*, ApJ **936** 172 (2022)
10. **A. M. Knee**, J. McIver, M. Cabero, *Prospects for measuring off-axis spins of binary black holes with Plus-era gravitational-wave detectors*, ApJ **928** 21 (2022)
11. **A. M. Knee**, D. Contreras, D. Scott, *Cosmological constraints on sterile neutrino oscillations from Planck*, JCAP **07** 039 (2019)
12. C. R. Mann, H. Richer, J. Heyl, J. Anderson, J. Kalirai, I. Caiazzo, S. D. Möhle, **A. Knee**, H. Baumgardt, *A*

multimass velocity dispersion model of 47 Tucanae indicates no evidence for an intermediate-mass black hole, ApJ **875** 1 (2019)

PUBLICATIONS – LARGE COLLABORATION

Selected publications from large collaborations, for which I note my contribution.

1. J. Glanzer *et al.* (inc. **A. M. Knee**), *LIGO detector characterization in the second & third parts of the fourth observing run*, submitted to CQG – I was on the paper writing team and contributed to the persistent data quality section
2. E. Goetz, A. Neunzert, **A. M. Knee** *et al.*, *Narrow spectral artifact investigation and mitigation in LIGO data from the fourth LIGO-Virgo-KAGRA observing run*, accepted in CQG – I was on the paper writing team and co-led the lines investigations effort
3. A. G. Abac *et al.* (inc. **A. M. Knee**), *Constraints on gravitational waves from the 2024 Vela pulsar glitch*, ApJ **1005** 12 (2026) – I was responsible for the HMM search pipeline and wrote the relevant section of the paper
4. S. Soni *et al.* (inc. **A. M. Knee**), *LIGO detector characterization in the first half of the fourth observing run*, CQG **42** 085016 (2025) – I documented spectral line artifacts in LIGO data and participated in data quality shifts
5. R. Abbott *et al.* (inc. **A. M. Knee**), *GWTC-3: Compact binary coalescences observed by LIGO and Virgo during the second part of the third observing run*, PRX **13** 041039 (2023) – I produced preliminary parameter estimation analyses for two gravitational-wave event triggers

TEACHING

Teaching assistant, University of British Columbia 2019 – 2024
Courses: PHYS 101, PHYS 403, ASTR 102, ASTR 310, ASTR 311

MENTORSHIP

Rishav Agrawal, Mitacs Globalink Internship, University of British Columbia 2024
Katja Nell, undergraduate thesis project, University of British Columbia 2022 – 2023
Neev Shah, Mitacs Globalink Internship, University of British Columbia 2022
Nikolas Boily, undergraduate thesis project, University of British Columbia 2021 – 2022

OUTREACH

Guest speaker, H. R. MacMillan Space Centre, *Vancouver, BC, Canada* Sep 2024
Public lecture on the science of stellar remnants
Lab volunteer, UBC Physics Olympics, *Vancouver, BC, Canada* Mar 2024
Lab assistant and judge
Mentor, UBC Launchpad Program, *Vancouver, BC, Canada* Jun 2022
Mentored prospective graduate students from historically underrepresented groups in physics and astronomy

SCIENTIFIC SERVICE

LIGO Scientific Collaboration
Liaison for Continuous Waves and Detector Characterization working groups (LSC org chart)
LIGO parameter estimation rota during O3 and O4 observing runs
LIGO data quality shifts during O4

Journals refereed
ApJ

PERSONAL

Citizenship: Canada
Languages: English (native), Spanish (fluent)